Software Project Documentation

Members

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Project Name: Unishop

1. Problem Definition

1.1 Introduction

In today's digital era, e-commerce platforms have become an essential part of business transactions worldwide. However, within the university community, student vendors often struggle to reach potential customers efficiently. The primary method of selling products and services is through door-to-door delivery, which is time-consuming and inefficient.

1.2 Problem Statement

Many students in Co-operative University of Kenya engage in business activities, selling products and offering services to their peers. However, these transactions are often informal and lack a structured platform. Key issues include:

- Difficulty in reaching potential customers.
- Lack of secure transaction methods.
- Inefficient order management and product visibility.

- The absence of a centralized platform for student vendors and customers to interact.

Without a structured e-commerce platform, student vendors find it difficult to scale their businesses, while potential customers struggle to find products and services efficiently.

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2. Problem Solution

2.1 Proposed Solution

To address the outlined challenges, we propose developing an e-commerce web application named **Unishop**. This platform will:

- Provide a secure and user-friendly online marketplace for students to buy and sell products/services.
- Enable vendors to create accounts, list their products/services, and manage orders.
- Allow customers to browse products, place orders, and make secure payments.
- Incorporate an automated payment system that sends a prompt to the customer's mobile number for transactions.
- Feature an integrated chat system for communication between administrators, vendors, and customers.

2.2 Features of Unishop

1. **User Authentication & Management:**

    - Students can sign up as either customers or vendors.
    - Vendors' products/services require approval from administrators.

2. **Product & Service Management:**
    - Vendors can add, edit, and delete their products/services.
    - Customers can browse, search, and favorite products/vendors.
3. **Order & Checkout System:**
    - Customers can add products/services to their cart.
    - Secure checkout process with integrated payment prompts.
4. **Chat Support System:**
    - Enables direct communication with administrators for assistance.
5. **Admin Dashboard:**
    - Manages vendors, customers, transactions, and product approvals.
6. **Scalability & Security:**
    - The system is designed for long-term usability with a focus on fast response time, security, and reliability.

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3. Methodology

3.1 System Development Life Cycle (SDLC)

The project will follow the **Agile Development Methodology**, ensuring continuous improvements based on user feedback. The key phases include:

3.1.1 Planning & Requirement Analysis

- Conduct surveys among students to identify core requirements.
- Define project scope, features, and functionality.

3.1.2 Design

- Develop wireframes and UI/UX designs.

- Create an entity-relationship diagram for database structuring.

3.1.3 Implementation

- **Frontend Development:** HTML, CSS, JavaScript.
- **Backend Development:** PHP for server-side logic.
- **Database Management:** MySQL for handling product, user, and transaction data.

3.1.4 Testing & Debugging

- Perform unit testing, integration testing, and user acceptance testing.
- Identify and resolve system bugs and performance issues.

3.1.5 Deployment

- Deploy the application on a local university server or cloud platform.
- Provide necessary documentation and training for end users.

3.1.6 Maintenance & Future Enhancements

- Regular updates to improve security and performance.
- Expand system functionalities based on user feedback.

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4. Conclusion

Unishop aims to revolutionize student commerce by providing a **structured, secure, and user-friendly** online trading platform. By integrating key features such as **secure payments, automated order processing, and real-time chat support**, the system ensures a seamless buying and selling experience for students. Through continuous development and user feedback, Unishop will evolve to better serve the university community's needs.